

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 1st Year 1st Term Regular Examination, 2019

URP1113
(Fundamentals of Planning Process)

Full Marks: 210

Time: 3.00hrs

- N.B.** i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.



Section – A

1. (a) "Planning is bringing the future into the present so that you can do something (10)
about it now."- How? Explain with an example in context of Khulna City.
(b) Explain the concept of Metropolitan City. How does urban sprawl cause (10)
environmental degradation during the growth of a Metropolitan City?
(c) "Planning has three dimensions in terms of Space, Sector and Time". Briefly (15)
explain all the three with relevant examples in the context of Bangladesh.
2. (a) You have been appointed as an intern in RAJUK (Rajdhani Unnayan (20)
Kartipakkha). You are assigned for the following assignment.
Dhaka City has experienced a number of devastating fire incidents in recent
months. The incidents killed/ injured many people and caused financial loss to
the nation. On the other hand, national media, experts and general people
blamed RAJUK for the incidents. RAJUK wants to take initiatives associated
with other Government agencies for minimizing the extent of the fire incidents.
Now you have to do the following tasks. Try to conceptualize your ideas in short.
Task 1: Prepare a Vision for solving the problem (please clarify your idea).
Task 2: What might be the Mission according to your Vision (please discuss
briefly).
Task 3: According to your Vision and Mission, formulate some Programmes.
Task 4: Mention some projects (i.e., your suggestion as project) under the one
Programme.
Task 5: Clarify Goal, Objectives and Strategies of any three projects among
your mentioned project.
(b) The lifecycle of urban development contains several phases. Briefly explain the (15)
phases with pertinent examples.
3. (a) A successful plan (in context of urban and regional planning) should be Specific, (20)
Measurable, Achievable, Realistic, Time Bound"- Why? Were the five
components followed during the present Dhaka Structure Plan?
(b) Do you support Citizen Participation in planning process? Draw the figure "A (15)
Ladder of Citizen Participation" and explain the types of participation and
nonparticipation with relevant examples.
4. Annotate any five of the following: (35)
(a) Project
(b) Programme
(c) Policy
(d) Strategic Plan
(e) Structure Plan
(f) Urban Area Plan
(g) Detailed Area Plan

Section – B

5. (a) Explain the role of Planners as negotiators in the modern era of urban planning. (10)
(b) Differentiate between Positivism and Post-positivism Paradigms of Planning with relevant examples. (10)
(c) Suppose a strong hurricane is approaching fast towards the coastal belt of Bangladesh and the Government has to take some immediate decisions to tackle the imminent disaster. Will a bottom-up approach be more effective than a top-down one in such a case? Why? Why not? (15)
6. (a) Draw a flow diagram showing the steps involved in a Comprehensive Planning Process. Mention the limitations of this approach. (15)
(b) Explain the concept of Mixed Scanning Approach of Planning with an appropriate example. (10)
(c) Write down the steps of Rational Planning with the help of an example (Hint: choosing the best transport mode for your city.) (10)
7. (a) Discuss the means of developing a Communication Strategy as part of the Advocacy Planning Process. (15)
(b) Write down and describe the core components of the Transactive Planning Process. (10)
(c) Explain the characteristics of Radical Planning Theory. (10)
8. (a) Explain the concept "SITAR" in planning theory. Provide a short overview of each planning approaches used in SITAR model. (15)
(b) Write short note on any 4 (four) of the following: (20)
 - (i) Top-down and Bottom-up Approach
 - (ii) Incremental Planning approach
 - (iii) Factory Towns
 - (iv) 3E's concept of sustainability
 - (v) Negotiator
 - (vi) Limitations and Strengths of Advocacy Planning

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 1st Year 1st Term Examination, 2019
URP 1111
(History/Evolution of Human Settlement)

Full Marks: 210

Time: 3 hrs

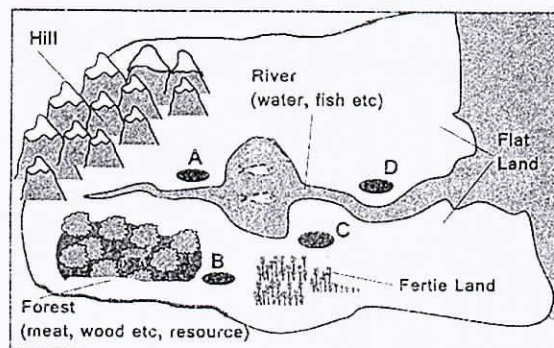
- N.B.** i) Answer any **three** questions from each section in separate script.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) Why should a planner need to learn the knowledge about “Evolution of Human settlement”? (10)
- (b) The “Primitive Organized” settlement is now converted into “The Universal City”. Briefly explain. (10)
- (c) The houses of new Stone Age people had a peculiar design. Why do you think so? (15)
2. (a) You had to choose a site for your settlement. What site selection factors would be important for your? (05)
- (b) The people of Bronze Age had some comparative advantage than the previous one. What are those issues? (10)
- (c) Write short notes on: (20)
(i) Pundravarddhan (ii) Shat Gambuj Mosque (iii) Vanga (iv) The Ring City.
3. (a) Developing countries have so many impediments in planning the rural settlements. What are those barriers? (05)
- (b) Classify towns based on their functional classification. (10)
- (c) Briefly discuss about the following settlements: (10)
(i) Rectangular pattern (ii) Star like pattern.
- (d) What is your understanding about “CBD” and “Primate City”? (10)
4. (a) The “Utopian thoughts” really paved the way of new thinking in city planning concept. Justify the statement. (10)
- (b) Briefly describe the concept of a Garden City. (15)
- (c) How did Le Corbusier plan to reduce the congestion of the center of the city? (10)

Section – B

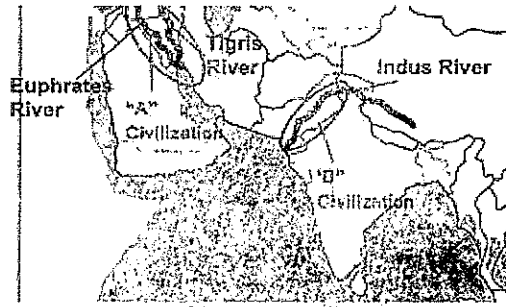
5. (a) Establish the relationship between site and situation of human settlement? Illustrate the “Doxiadis Framework” of human settlement. (10)
- (b) Imagine that you and your tribe have just arrived in a place of the figure below. As a Team Leader choose the best site among A,B,C,D for your village by the scoring method. (15)



- C) “Mesopotamian civilization was in favor of civic facilities”. Justify your answer. (10)



6. a) What are the "A" and "B" civilization in the above figure? Write down some major features of the both civilization. (15)



- (b) Nile river is Known as the "Giver of Life" during Egyptian civilization. Clarify the statement with necessary sketch. (10)
- (c) Show the pyramid evaluation and set back rule of Egyptian civilization with necessary sketch. (10)
7. (a) Differentiate between the "Parthenon" and "Pantheon" with necessary sketch. (10)
- (b) Discuss about one of the ancient public places of ancient Greek and Roman civilization. (12)
- (c) Differentiate among the Roman social classes. (07)
- (d) How the feudalism social structure of Mesopotemian civilization interconnected? (06)
8. (a) 'Hippodamus' grid iron pattern was really beneficial- Do you think so? Justify your answer. (08)
- (b) Define the concept of "Dark Age"? Discuss about the revolution of transport after renaissance period. (15)
- (c) Describe about the water supply system above the river to the ancient Rome with necessary sketch. (07)
- (d) Write down the five characteristics of Hellenistic city. (05)

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 1st Year 1st Term Examination, 2019

CH 1117
(Environmental Chemistry)

Full Marks: 210

Time: 3.00 hrs

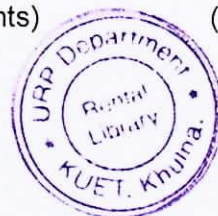
- N.B. i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) Draw the molecular orbital diagram of following: (10)
(i) HI (ii) Cl₂ (iii) He₂⁺
and find out their bond order and magnetism.
(b) Discuss the nature of hybridization, shapes and bond angle of CH₃CH₃, H₂O and (10)
NH₃ molecules.
(c) Draw potential energy diagram for the formation of a diatomic molecule as a (08)
function of intermolecular distance.
(d) What is hydrogen bonding? How does intermolecular and intramolecular (07)
hydrogen bonding influence the physical properties of the compounds?
2. (a) Draw a thematic diagram of a nuclear power plant and briefly explain its different (10)
components.
(b) Define mass defect. Deduce the relationship between packing fraction and (10)
nuclear stability.
(c) How many α - particles per second would be emitted from 4×10⁻¹² g of ²¹⁰Po (07)
(*t*_{1/2} = 138 days).
(d) Write few industrial and biological applications of radio-isotopes. (08)
3. (a) "Transition metals have high melting and high boiling point compared to alkali (10)
and alkaline earth metals' – explain in favor of your answer.
(b) What is meant by molarity and molality of solution? Which of these two is (10)
temperature dependent?
(c) Calculate the molality of a solution of sodium hydroxide which contains 0.2 g of (07)
sodium hydroxide in 50 g of the solvent.
(d) Write down different types of solutions along with examples. (08)
4. (a) What is meant by 'buffer solution' and 'buffer action'? Explain clearly why a (10)
solution of a weak acid and its salt with a strong base behaves as a buffer
solution.
(b) Classify the following substances as acids and bases, giving reason: BF₃, NH₄⁺, (10)
CaO, Cl₂ and Na.
(c) Calculate the pH of 0.001N HCl assuming complete ionization of HCl. (07)
(d) Discuss briefly the properties of transition elements (d-block elements) (08)

Section – B

5. (a) What do you mean by the terms: (i) Substrate (ii) Attacking reagent and (iii) (10)
Intermediates? Apply these terms to a substitution reaction.
(b) Why tertiary Carbonium ions are more stable than primary Carbonium ions? (09)



- (c) Arrange the compounds of each set in order of reactivity towards S_N1 and S_N2 reaction: (08)
- (i) 2-bromo-2-methylbutane, 1-bromopentane and 2-bromopentane.
 - (ii) 1-bromo-3-methylbutane, 2-bromo-2-methylbutane and 2-bromo-3-methylbutane.
- (d) What is meant by racemization and inversion of product? (08)
6. (a) What is polymerization? Classify polymers with different point of view. (10)
- (b) What is stereopolymer? Briefly discuss the properties of isotactic, syndiotactic and atactic polypropylene polymer. (15)
- (c) Discuss the reaction mechanism of free radical polymerization. (10)
7. (a) Graphically represent the major regions of the atmosphere with temperature profile and write the important chemical species in each step. (10)
- (b) What is greenhouse effect? What are the consequences of the depletion of ozone layer? (07)
- (c) Write down the characteristic properties of wastewater. Discuss briefly the treatment process of domestic wastewater. (10)
- (d) Define industrial hazards. Write down the different types of chemical hazards with example. (08)
8. (a) What is meant by corrosion? Discuss the factors that enhance the rate of corrosion in Khulna region. (08)
- (b) Discuss the electro-chemical mechanism of wet corrosion. (08)
- (c) What do you mean by atmospheric corrosion? Write down the effect of humidity and dust particles on the rate of corrosion. (10)
- (d) Discuss the prevention of corrosion by modification of environment. (09)

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Hum 1117

(Functional English)

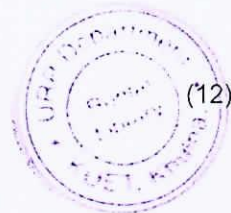
Full Marks: 210

Time: 3 hrs.

- N.B.** i) Answer any three questions from each section in separate script.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) Make sentences on the following structures. (14)
- (i) Subject + Intransitive Verb + Adverbial + Extension
 - (ii) Subject + Linking Verb + Adjective Complement + Extension
 - (iii) Subject + Linking Verb + Noun Complement + Extension
 - (iv) Subject + Transitive Verb + Gerund as Object
 - (v) Subject + Transitive Verb + Infinitive as Object
 - (vi) Subject + Transitive Verb + Object + Adjective Complement
 - (vii) Subject + Be Verb + Adverbial
- (b) Change the following words as directed and make sentences with the changed words: Free (into verb); Money (into adjective); People (into adj.); People (into verb); Necessary (into noun); Necessary (into verb); Central (into adj.) (12)
- (c) Make new word with the following prefixes and use the new words in sentence. (09)
- Auto _____, Bi _____, Contra _____, Mega _____, Sub _____, Tri _____
2. (a) Frame wh questions from the underlined parts of the following : (14)
- (i) The shop at the corner is mine.
 - (ii) The train leaves at 7.30 am.
 - (iii) Death is preferable to dishonor.
 - (iv) Rajshahi is famous for mango.
 - (v) He wanted to sing.
 - (vi) He lives in Khulna.
 - (vii) Ha has earned 1000 /-.
- (b) Make sentences using the following words as directed. Right (as noun); Right (as verb); Head (as verb); Head (as adj.); Stage (as verb); Home (as verb). (12)
- (c) Write two antonyms for each of the following words and make sentence with the antonyms. (09)
- Cheerful, Passion, Generous
3. (a) Transform the following sentences as directed. (14)
- (i) Nothing is difficult there. (Interrogative)
 - (ii) It is true that he may get a scholarship. (Passive)
 - (iii) There are not many students in the college. (Affirmative)
 - (iv) We know him to be honest. (Complex)
 - (v) It was simply a rumor. (Negative)
 - (vi) If you can try heart and soul, you can fulfill your dream. (Simple)
 - (vii) Having heard the news, he was shocked. (Compound)
- (b) Make use of the following modals in sentence as asked in brackets. (12)
- (i) Can (To express offer)
 - (ii) Will (To express determination)
 - (iii) Would rather (To express preference)
 - (iv) Should (To express an action which was not discharged)
 - (v) Would (To express a polite request)
 - (vi) Used to (To express a regular habit in the past)
- (c) Make sentences using the following phrases and idioms. (09)
- At a time; Far and wide; Hue and Cry; In a fix; Out of date; Rank and file



4. (a) Correct the following sentences. (14)
 (i) His absence was owing to illness; (ii) My father prevented me to out at night; (iii) Would you mind to wait a moment? ; (iv) Neither he nor his brother has not done this; (v) He resigned from the post; (vi) He has come home yesterday; (vi) Labu plays piano.
- (b) Express the following notions /functions in sentence. (12)
 (i) Invitation; (ii) Determination; (iii) Intention; (iv) Regrets; (v) Good Luck; (vi) Command
- (c) Identify the parts of speech of the underlined words of the following sentences. (09)
 (i) All that glitters is not gold; (ii) The fishermen near the shore; (iii) Come near; (iv) He is a near relation of mine. (v) I saw the man near the station; (vi) It's your round now.

Section – B

5. (a) Read the following passage carefully and answer the questions that follow. (20)
- Natural hazards affect our life so much frequently, but after they occur we forget what to do. Can we forget the concept – "Prevention is better than cure?" Every year more or less we face flood and cyclone, and catastrophically we fear the threat of global warming. For these hazards we and the industrialized countries of the world are equally responsible. Flood and cyclone destroy severely our economy. Crops, homes, roads, different constructions etc. face a great havoc. We temporarily recover the loss, they do, but our country both the authority and the peoples, should equally come forward to save permanently our country from the threat of flood and cyclone. For this purpose what we should take action indicates to define the reasons – why flood and cyclone happen, how we can remove them permanently etc. In so way that water of rainfall can take shelter easily without creating flood, we should plant more trees so that cyclone can't create calamity, we should ensure the flow of the water of a river by dragging its bed, we should negotiate with the partners of international rivers to save from the overflow of water especially in the rainy season etc. Another hazard, which the industrialized countries and also the countries using more vehicles create, is the global warming and other hygienic problems. The industrialized countries and the countries using vehicles more produce goods for our necessity and use more vehicles in roads, but are not aware of the consequence of global warming. This global warming is responsible for the increase of carbon dioxide, which increases the temperature of the earth atmosphere. As a result the earth loses balance of the temperature for the overall development of all types of plants, animals etc. And icy area of the world loses frozen state and commonly sea level rises up. And the countries adjacent to sea, are at a great risk. The earth is for human beings. Any one, responsible for the loss of balance of ecosystem, represents himself as an enemy to the people of world.
- Questions:
- (i) How do flood and cyclone destroy our economy?
 (ii) What should we do to save our country permanently from the danger of flood and cyclone?
 (iii) What type of actions should we do against flood and cyclone?
 (iv) What does the global warming affect?
- (b) Make a précis of the above written passage (Q.5.a) with a suitable title. (15)
6. (a) Amplify the idea "Industry is the mother of good luck". (20)
 (b) Write a paragraph on Patriotism. (15)
7. (a) Write a report on Environmental Pollution – You suffer around your hall. (20)
 (b) Write a letter to the editor of a newspaper about widespread practice of taking bribe. (15)
8. Write a free composition on any one of the following : (35)
 (a) Planned City : A Demand of the day
 (b) Agriculture and the Economy of Bangladesh

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP1st Year 1st Term Regular Examination, 2019

Math 1117
(Mathematics - I)

Full Marks: 210

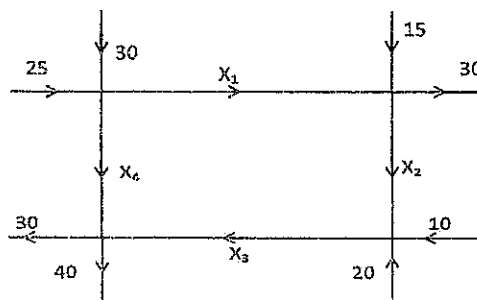
Time: 3.00 hrs

- N.B.** i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.
iii) Necessary charts and graphs will be supplied.

Section – A

1. (a) Establish a relation between circular and hyperbolic function. (10)
- (b) A town is built to accommodate maximum of 30,000 persons. At the starting year there was 1,000 habitants. At the end of the 1st year there was 2,500 inhabitants. What will be the population model? When the population of the town be 20,000 ? (15)
- (c) Two planes are flying in such a manner that they always remain at the same meridian. Let plane A is flying at the 30° latitudinal circle with speed 500 Knots. If plane B flies at 60° latitudinal circle then what should be its speed? (10)
2. (a) For any spherical triangle prove that, (12)

$$\sin a \cot b = \cos a \cos C + \sin C \cot B$$
- (b) Solve the quadrantal triangle for which $c=90^\circ$, $a=105^\circ 25'$ and $B=128^\circ 35'$ (12)
- (c) Find the shortest distance between two places having locations (35°25'N, 105°35'W) and (37°15'S, 116°40'E) (11)
3. (a) Define skew symmetric matrix. Show that the diagonal elements of a skew-symmetric matrix are zero. (11)
- (b) Reduce $A = \begin{bmatrix} 1 & 2 & 3 & -2 \\ 2 & -2 & 1 & 3 \\ 3 & 0 & 4 & 1 \end{bmatrix}$ to normal form and also find its rank. (12)
- (c) Define elementary matrix. Show that the effect of pre-multiplication by E_{12} is nothing but interchange between row 1 and 2. (12)
4. (a) When does a matrix have its inverse? If possible find the inverse (11)
of $\begin{bmatrix} 2 & 3 & 4 \\ 3 & 5 & 7 \\ 1 & 7 & 3 \end{bmatrix}$
- (b) Find the rank of $A = \begin{bmatrix} 2 & 3 & -1 & 4 \\ 6 & 9 & -3 & 12 \\ 4 & 6 & -2 & 8 \end{bmatrix}$ (06)
- (c) In the following traffic system can we block the route x_4 ? If 40 vehicles are permitted to pass through route x_4 , what will be volume of traffic in other routes? (18)



Section – B

5. (a) The equation $3x^2 + 2xy + 3y^2 - 18x - 22y + 50 = 0$ is transformed to $4x^2 + 2y^2 = 1$ when referred to rectangular axes through the point (2,3). Find the inclination of the later axes to the former. (12)
- (b) Transfer $4x^2 - 16xy + 9y^2 - 7x + 5y - 9 = 0$ to its standard form. Hence (18)
find the length and equations of its axes.
- (c) What will happen if the origin is shifted at the centre of a conic or on the conic? (05)
6. (a) Derive the relation between Cartesian and spherical polar coordination (Figure (15)
is necessary). Obtain the spherical polar coordinates of $(-2, 2\sqrt{3}, -4\sqrt{3})$.
- (b) Determine the equation of the parabola $x^2 - 2xy + y^2 + 2x - 4y + 3 = 0$ (10)
after rotating of axes through 45° .
- (c) Test the nature of the conic given by the equation $3x^2 - 8xy - 3y^2 + 10x - 13y + 8 = 0$. (10)
7. (a) Find the equation of the plane through the points (2, 2, 1) and (9, 3, 6) and (11)
perpendicular to the plane $2x + 6y + 6z = 9$.
- (b) Find the distance of (2, 3, -4) from the plane $2x + 4y - 5z + 7 = 0$ measured (12)
parallel to the line $\frac{x}{-1} = \frac{y}{2} = \frac{z}{3}$.
- (c) Two system of rectangular axes have the same origin. If a plane cuts the axes (12)
at a, b, c and a_1, b_1, c_1 respectively from the origin. Prove that,
- $$\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} = \frac{1}{a_1^2} + \frac{1}{b_1^2} + \frac{1}{c_1^2}$$
8. (a) Find the angle between the line $2x + 4y - 2z + 3 = 0 = 4x - 2y + 6z + 5$ (10)
and the plane $5x - 4y + 3z - 5 = 0$
- (b) Find the magnitude and the equations of the line of shortest distance between (14)
the lines $\frac{x-3}{3} = \frac{y-8}{-1} = \frac{z-3}{1}$; $\frac{x+3}{-3} = \frac{y+7}{2} = \frac{z-6}{4}$.
- (c) Show that the lines $\frac{x+3}{2} = \frac{y+15}{3} = \frac{z-7}{-3}$; $\frac{x+1}{4} = \frac{y+1}{5} = \frac{z+1}{-1}$ are (11)
coplanar and find the equation of the plane containing them.